Safeguarded AI-Driven Semantic Communication: Design Principles, Architecture, and Challenges

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Abstract

Semantic communication (SemCom) systems heavily rely on artificial intelligence (AI) models to generate semantic representations and reconstruct original meaning. However, current AI models in SemCom systems present reliability concerns due to two major challenges. First, they often operate on a "best effort" basis, displaying unwarranted overconfidence that undermines fidelity in semantic reconstruction. Second, they are lack of generalization capabilities to adapt to changing network conditions and varying knowledge bases (KBs) in SemCom networks. Communication systems are safety-critical, which cannot tolerate the unreliable transmission caused by unsafe AI behavior. In this paper, we establish a safeguarded AI framework for SemCom networks that constrains AI behavior through defined safety mechanisms and boundaries to address the two challenges mentioned above. Particularly, we present safeguarded AI's architecture, detailing the designs of world model, safety specifications, and gatekeepers. Notably, gatekeepers enforce safety constraints on AI output based on defined safety specifications. We also implement a real-world safeguarded AI-driven SemCom framework with periodically updating KBs to absorb new environmental data and users' feedback. Furthermore, we validate our framework's effectiveness through a case study with simulation results. Finally, we explore additional challenges of the proposed framework to guide future research directions.

Introduction

Semantic communication (SemCom) has been deemed as a promising paradigm that focuses on conveying the semantic meaning embedded within messages [1]. The system works through semantic encoders and decoders equipped with background knowledge, enabling accurate meaning interpretation even in poor communication environments.

SemCom systems are inherently safety-critical. This criticality arises partly from the foundational role that communication systems play in enabling other safety-critical infrastructures, such as electrical power grids, intelligent transportation systems, air traffic control, autonomous vehicle networks, and industrial automation. These domains demand ultra-reliable and low-latency communication (URLLC) to ensure the timely and accurate exchange of information, where even slight communication errors can lead to severe consequences. SenCom, underpinned by AI models, has emerged as a key enabling technology to meet the stringent URLLC requirements. Unlike conventional communication systems that transmit syntactic symbols, SemCom systems transmit task-relevant semantic representations, which introduces new safety considerations.

The semantic encoder and decoder must jointly ensure robustness and safety, as errors introduced during compression, transmission, or reconstruction can lead to semantic inconsistencies that traditional error control mechanisms cannot easily detect or correct. Even minor perturbations in semantic representations can result in significant misinterpretations, making end-to-end integrity critical in safety-sensitive applications.

Current AI models provide superior semantic processing tools for SemCom systems through perfect training in ideal and pre-defined communication conditions. However, as illustrated in Fig. 1, in real-world deployment, AI models in SemCom networks face unforeseen network dynamics and unseen message input that expose critical safety vulnerabilities. Especially, these models frequently lack rigorous safeguards to address safety concerns in three fundamental aspects:

- AI models operate on a "best effort" basis, such as maximum likelihood estimation, which means that models do their best to produce an output, regardless of whether it's reliable. Even when confronted with previously unseen inputs, these models still generate outputs without guarantees. It is crucial when these models are adopted as semantic encoders, as it can result in inaccurate semantic information, further affecting the data reconstruction in semantic decoder.
- Sometimes AI models are overconfident during prediction, assigning high confidence scores to incorrect outputs. Since

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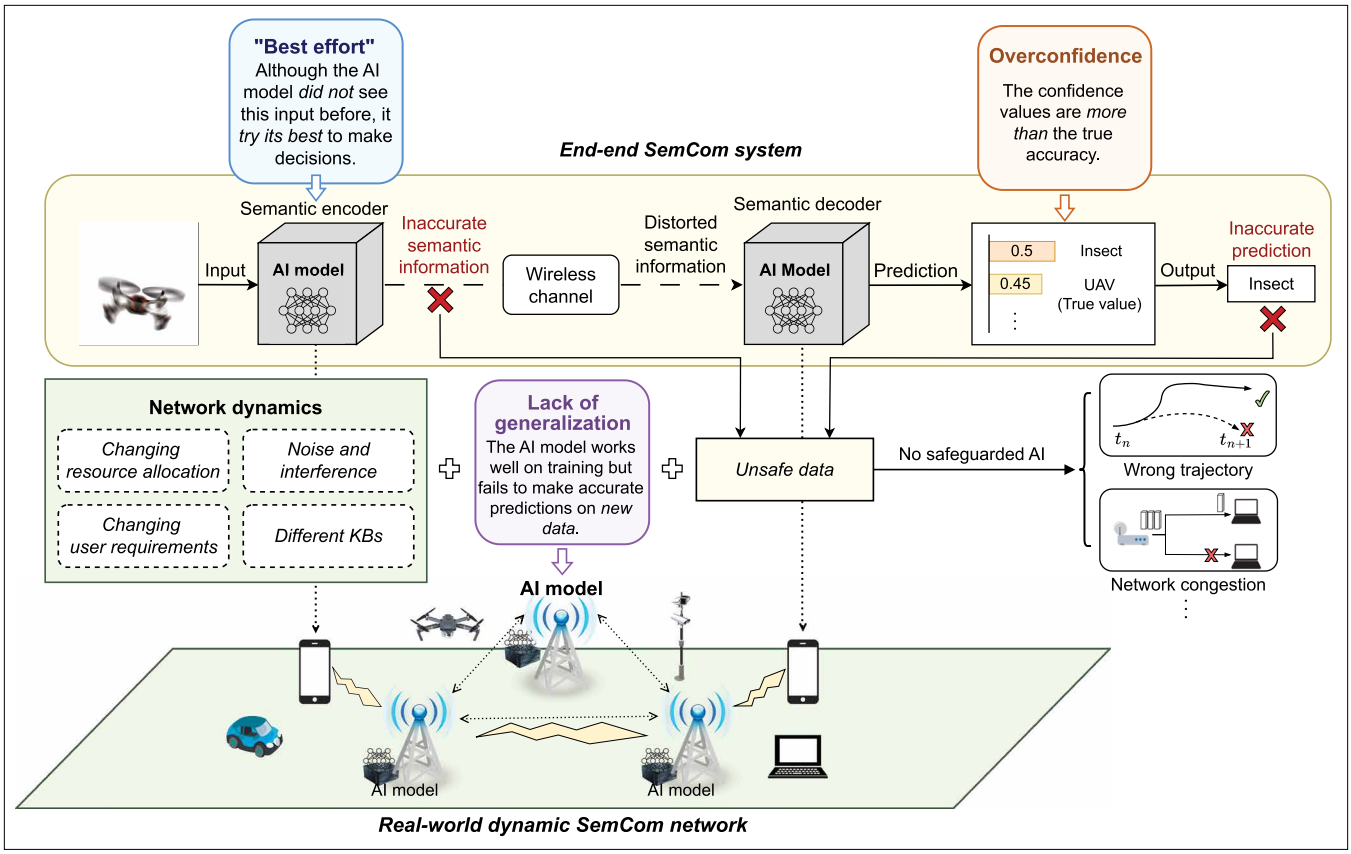

FIGURE 1. Three problems of existing AI models in SemCom networks. The upper layer demonstrates end-to-end semantic transmission, where problems of "best effort" and overconfidence arise. The bottom layer represents a real-world dynamic SemCom network. The combination of network dynamics and the inaccurate output generated at the upper layer manifests as the "lack of generalization" problem.

these models typically make decisions solely based on the highest confidence scores, in the semantic decoder, this overconfidence manifest *s* as inaccurate message reconstruction.

- Network dynamics, encompassing changing channel conditions, resource allocation, and evolving user requirements, are frequent and inherent to SemCom networks. However, many AI models lack the generalization capabilities to adapt to these dynamic conditions. Under such network dynamics, the problems of "best effort" basis and overconfidence become particularly severe. Additionally, the limited generalization capabilities create specific challenges when dealing with diverse knowledge bases (KBs) in SemCom networks, which is critical to semantic encoding and decoding.

To address these challenges, the concept of "*safeguarded AI* " is introduced. Safeguarded AI is a framework aimed at ensuring that AI systems operate under strict safety protocols, especially in safety-critical applications. Building upon the broader goals of AI safety, which seeks to create systems that are human-centered, controllable, interpretable, robust, and aligned with human values, safeguarded AI distinguishes itself by employing mathematical rigor and formal verification methods over empirical safety testing to enforce these safety objectives. The framework's core innovation is an external *gatekeeper* module that provides mathematical guarantees for keeping AI outputs within proven safety boundaries [2]. Rather than relying on the model's internal confidence scores, which are often prone to overconfidence or "best-effort" decisions, the gatekeeper performs independent assessments using separate safety criteria. It acts as a safety layer that monitors real-world interactions and enforces predefined safety boundaries tailored to the system's intended use. To enable adaptive evaluation of AI behavior under dynamic conditions, the gatekeeper continuously collects network condition parameters and simulates prediction distributions. Furthermore, it can feed the collected real-time data back into the AI pipeline, enriching the training dataset and thereby enhancing the model's robustness and generalization performance over time.

Although there is no existing work leveraging safeguarded AI in SemCom networks, recent advances in AI have brought AI safety concerns to the forefront of research attention. The authors [3] propose a "safeguarded AI" framework, envisioning collaboration between humans and frontier AI systems. Fundamental properties of safe AI systems have been extensively explored in [4] and [5], which also investigate formal methods for verifying these properties. Building on this foundation, the authors [3] introduce the concept of "guaranteed safe AI", establishing a systematic family of approaches to AI safety. Also, the authors in [6] analyze safety challenges and mitigation strategies for large language models. Moreover, some pioneering work address AI

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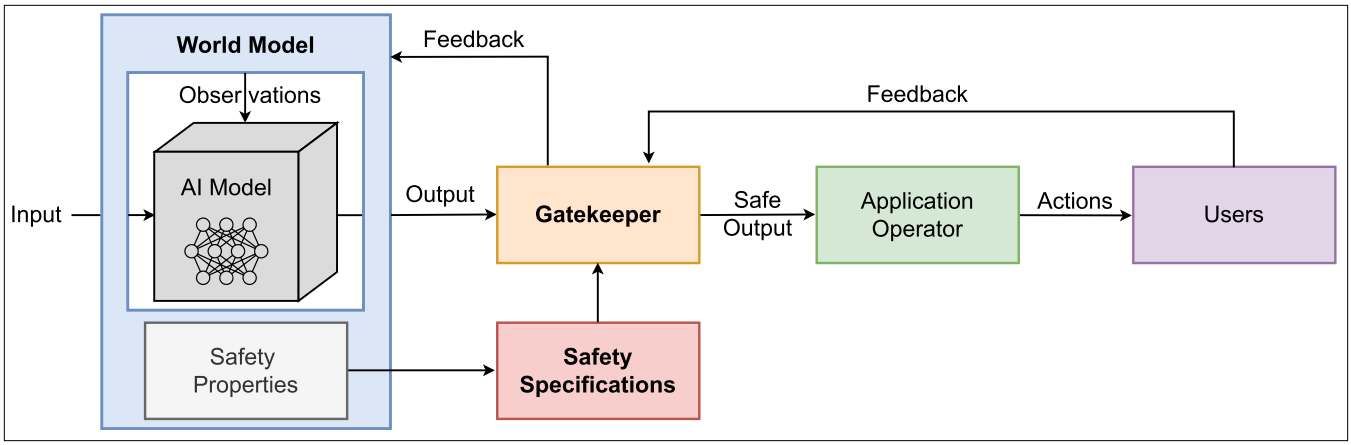

FIGURE 2. The architecture of safeguarded AI framework.

safety in communication scenarios. The authors in [7] present a comprehensive survey of AI-based safety solutions across various communication technologies and application domains. The authors in [8] advance this field by enhancing AI model safety through conformal prediction techniques with formal calibration guarantees.

In contrast to existing work, we guarantee the safety of SemCom systems through the development of a safeguarded AI framework. We begin by presenting the architecture of the safeguarded AI framework with three major components: world model, safety specifications, and gatekeepers. We explain the design principles of these three components in SemCom systems. Next, we propose a safeguarded AI-driven SemCom framework and introduce its architecture and implementation. Simulation results from a case study demonstrate significant improvements in semantic accuracy compared to conventional SemCom methods.

Safeguarded AI Design Principles

In this section, we introduce the design principles in safeguarded AI framework. Fig. 2 illustrates the architecture of safeguarded AI framework. An AI model operates within a world model that simulates or represents the operational environment. The model generates outputs based on its inputs, which are then evaluated by a gatekeeper according to safety specifications derived from the world model. Only outputs deemed safe proceed to the operator, who is responsible for executing these approved actions in the real-world system. Finally, users, such as domain experts, monitor the outcomes and behaviors post-execution and provide feedback to gatekeepers, allowing future safety evaluations and updates. Specifically, the world model, safety specifications, and gatekeeper are three main components:

- *World Model:* A world model serves as a mathematical and logical representation of real-world dynamics, establishing the formal foundation necessary for rigorous safety verification in safeguarded AI framework. The world model transforms abstract safety concepts into verifiable constraints by converting vague directives into precise mathematical inequalities and logical predicates over defined model states. By systematically capturing safety-relevant environmental aspects, the model provides structured observations that enable the AI model's predictive reasoning about potential actions and safety implications.
- *Safety Specifications:* Safety specifications contain the safety properties of AI systems, defining the conditions under which the system is considered to operate safely. These specifications typically include constraints on system behavior, acceptable operating ranges, failure modes to avoid, and responses to uncertain or adversarial inputs. They serve as formal criteria for verifying and validating the system's reliability, robustness, and compliance with safety standards throughout its deployment lifecycle.
- *Gatekeepers:* The central concept of safeguarded AI is gatekeepers, which function as an oversight mechanism for autonomous AI systems. A gatekeeper defines the safety boundaries and ensures that the outputs and actions of AI systems strictly adhere to predefined and verifiable safety boundaries. According to this model, an AI output is deemed safe only if it remains within these boundaries. Advanced methodologies, including mathematical analysis and machine learning (ML) techniques, are employed to monitor, assess, and certify the safety of AI decisions in real-time.

However, narrowing down to SemCom, the design of these three components presents substantial challenges.

How to build world models for SemCom networks? The world model for SemCom networks is hybrid, integrating both syntactic and semantic levels. The syntactic level encompasses traditional communication models, such as channel models. In contrast, the semantic level involves models for semantic encoding and decoding, as well as sematic aware adaption. The hybrid model introduces new aspects to traditional problem formulations, such as semantic-aware resource allocation strategies. Consequently, building an world model for SemCom networks is a complex challenge, requiring the seamless integration of syntactic and semantic components while addressing the dynamic nature of communication environments.

How to define semantic safety specifications? Safety specifications in SemCom networks involve abstract and context-dependent

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factors. In this context, the definition of "safety" is extended beyond conventional reliability measures, including the accuracy and consistency of semantic understanding. For instance, a mismatch in interpretation between different models can be considered "unsafe", potentially leading to miscommunication and unintended actions. However, defining and measuring semantic safety remains ambiguous.

How to design gatekeepers in SemCom networks? In SemCom networks, gatekeepers should operate in real-time while monitoring complex semantic interpretations, requiring sophisticated algorithms that balance computational efficiency with monitoring accuracy. Moreover, gatekeepers need to adapt to varying network conditions and semantic contexts without compromising its monitoring reliability.

Safeguarded-AI in Semantic Communication Networks

To solve the challenges discussed in the "Safeguarded AI Design Principles" section, in this section, we investigate potential approaches to world model construction, safety specification formulation, and gatekeeper design in SemCom networks.

World Models

In order to explore the safety boundaries for SemCom networks, we should first develop a world model to describe the environment in which SemCom models operate. In particular, the tasks in SemCom networks are conceptualized at two levels: end-to-end level and network level. First, at the end-to-end level, we consider the encoding and decoding processes of semantic representations as target AI model. The external changing factors, i.e., semantic representation methods, KBs, and channel conditions, constitute the world model. Second, the tasks at the network level encompass bandwidth allocation, knowledge updates, power control, etc. Here, decision-making AI models serve as the target models which are responsible for analyzing network conditions and implementing optimal decision strategies. The environment elements of these models are related to time-varying channel conditions, user allocation, and multidimensional resource constraints. Notably, knowledge resources are essential in SemCom networks, as they directly impact the accuracy of semantic encoding and decoding processes. Meanwhile, knowledge processing need significant memory and computing capabilities, which requires robust models and efficient knowledge allocation strategies under resource constraints.

To generally describe the environment of SemCom networks, we investigate various approaches that leverage both mathematical tools and ML techniques.

1. *Mathematical Tools:* For diverse goals and environment in SemCom networks, there are four categories of mathematical tools:
   - Random Spatial Patterns: *Stochastic geometry* are commonly used to model the randomness SemCom networks in both temporal and spatial domains. For example, Poisson-based models are usually utilized to derive the coverage probability of heterogeneous cellular network. In SemCom networks, tractable analysis of semantic-aware metrics can be derived by exploiting these models, such as KB synchronization rates between distributed nodes and maximum semantic information rate achievable under given channel conditions.
   - Probabilistic Inference: *Probabilistic models*, such as Bayesian networks, are capable to capture conditional dependencies among multiple variables. A Bayesian network is a directed acyclic graph $G = (V,E)$, where each node $v \in V$ represents a random variable, each directed edge $(u \rightarrow v) \in E$ represents a dependency between variables, and the joint probability distribution $P(V)$ factorizes according to G as $P(V) = \prod_{v \in V} P(v \mid \mathrm{Pa}(v))$, where $\mathrm{Pa}(v)$ denotes the parent nodes of v in G. Specifically, the intricate probabilistic relationships between semantic content, knowledge states, and SemCom network conditions can be captured by such probabilistic models.
   - Random Process: *Markov models* [9] and stochastic processes emphasize sequential structure, focusing on transitions between states to efficiently capture temporal dynamics in modeling. A Markov chain is defined by a set of states $S = \{s_0, s_1, ..., s_n\}$ and a set of directed transitions between these states. The process begins at an initial state s_{ini}, and the evolution of the model is governed by transition probabilities p_{ij}, where p_{ij} denotes the probability of transitioning from state s_i to state s_j. In Markov models, the next state depends only on the current state and not on the sequence of states preceding it. Particularly, these models are particularly well-suited for dynamic semantic channel modeling, SemCom network state prediction, semantic encoder and decoder evolution, and KB updates.
   - Stochastic Decisions: for scenarios involving multiple agents, *game theory* model how agents choose strategies to maximize their utility, considering the actions of others. Equilibrium concepts (e.g., Nash equilibrium) are used to predict how players act. For instance, a possible strategy for player i is denoted as s_i, where $i = 1, ..., N$. A strategy profile, a set consisting of one strategy for each player, is represented as $s^* = \left(s_i^*, s_{-i}^*\right)$, where s_{-i}^* denotes the $N - 1$ strategies of all the players except i. Additionally, the player i's payoff is $u_i\left(s_i, s_{-i}^*\right)$ as a function of the strategies. In SemCom networks, it can be utilized in competitive or cooperative tasks, e.g., resource allocation, multiple access control, network security, and knowledge sharing.
2. *Machine Learning Tools:* Compared to mathematical tools, ML approaches offer greater flexibility and intelligence for more complex environment, though they show

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less interpretability. ML tools for SemCom can be classified into three main categories: deep learning (DL), reinforcement learning (RL), and causal learning (CL). Notably, while DL provides the foundation, RL and CL are particularly valuable for modeling the evolution and adaptation of SemCom systems over time.

- DL serves as a fundamental modeling approach, particularly for end-to-end SemCom systems. These deep neural network models excel at extracting and encoding source data and knowledge into semantic representations, as well as interpreting semantic representations back into source data. Also, instead of manually defining physical or communication dynamics, these networks are trained on observed data to predict environmental changes, channel variations, or user behaviors over time.
- RL is widely used to learn the environment's dynamics, specifically how the world model state evolves in response to actions. During interactions with the environment, the agent collects experience tuples (s_t, a_t, s_{t+1}, r_t), consisting of the current network state, action taken, resulting next network state, and received reward. These experience samples are then used to train a RL network that models the environment's behavior. The learned model includes a state transition function $s_{t+1} = f(s_t, a_t)$, which predicts the next state given the current state and action, and a reward function $r_t = g(s_t, a_t)$, which predicts the immediate reward associated with the state-action pair. In SemCom networks, the model state specifically encompasses semantic encoder/decoder parameters, KB updates and consistency metrics, contextual understanding levels, and semantic channel conditions. The reward design is tailored to semantic-aware metrics for various tasks, such as semantic fidelity preservation, KB alignment across nodes, and computational efficiency.
- CL can capture cause-and-effect relationships between semantic concepts, KBs and network states though stable causal mechanisms. It improves the interpretability of SemCom model outputs and generalization across different contexts, KBs and network conditions. As one of the core CL methods, a structural causal model (SCM) can be formally represented as a tuple $M(\mathbf{U},\mathbf{V},\mathbf{F},P(\mathbf{U}))$, where U,V represent the sets of external and model variables, respectively. The set $F = \{f_1, f_2, ..., f_n\}$ defines deterministic functions such that each $V_i \in \mathbf{V}$ is determined by $V_i = f_i(\mathrm{Pa}(V_i), U_i)$. These causal models can be integrated into Bayesian networks for semantic inference and RL frameworks for adaptive SemCom strategies.

Safety Specifications

Safety specifications identify safety properties derived from the world model, serving as objective functions and/or constraints in AI models for SemCom networks. These specifications are classified into four categories.

- The scale of *knowledge domains* in KBs and the *KB alignment coefficient* between transceivers represent critical safety specifications. These metrics must exceed defined thresholds to ensure the accuracy of semantic encoders and decoders. Specifically, insufficient knowledge domain coverage will result in inaccurate semantic interpretation. When the KB alignment coefficient falls below the established minimum threshold, knowledge sharing protocols between transceivers should be triggered to achieve proper alignment.
- *Semantic effectiveness* is fundamentally connected to AI safety because it measures whether SemCom models accurately preserve and convey intended meaning. This effectiveness can be assessed through metrics such as semantic similarity calculated at the semantic decoder output.
- The *semantic quality of service (S-QoS)* focuses on the meaning preservation, in contrast to traditional QoS. For example, it quantifies how well semantic information is preserved relative to bit consumption and semantic transmission latency [10]. Degraded S-QoS may signal that SemCom models cannot process semantic information quickly enough for safe real-time decisions.
- *Age of Information/Incorrect Information (AoI/AoII)* in SemCom measures how timely information and knowledge reaches the receiver compared to when it was generated, or how long incorrect information and knowledge persists before correction. When AoI or AoII values are high, SemCom models risk making unsafe decisions based on outdated or incorrect semantic information. Thus, low AoI and AoII are critical safety requirements for time-sensitive applications.

Gatekeeper Design

According to safety specifications, gatekeepers are implemented to develop and maintain decision-support tools for the output generated by AI models in SemCom networks. As illustrated in Fig. 2, the gatekeeper is positioned between the AI model and application operator. It analyzes feedback received from users to determine the appropriate safety boundaries. It then filters out any outputs that fall outside the defined safe region and forwards only the reliable outputs to the operator.

Let us discuss four techniques to design and implement gatekeepers in SemCom systems.

- Simple *filter* based on predefined safety specifications is a fundamental mechanism in gatekeepers, which strictly maintain AI behavior within established safety boundaries. These safety boundaries are established through rigorous empirical testing and calibrated to meet diverse user requirements. Moreover, gatekeepers can evolute by dynamically adjusting these boundaries based on ongoing performance analysis and evolving user needs.
- *Random smoothing* can be developed to analyze the robustness of AI models in

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SemCom networks. This method works by creating a smoothed classifier for semantic encoding/decoding around a its model by adding calibrated Gaussian noise to inputs before classification. By aggregating predictions across multiple noise-perturbed versions of the same input, random smoothing establishes probabilistic guarantees of consistent classification for semantic information within a specified radius around any input point.

- *Conformal prediction* [11] is a method that transforms the output of SemCom model into a prediction set instead of a single prediction. Therefore, it can be used to predict the original data in semantic decoding process. In particular, this method offers formal guarantees of reliability that satisfies P (true output $\in$ predicted set) $\geq 1 - \alpha$, for any user-defined miscoverage level α. Moreover, conformal prediction adaptively adjusts its thresholds based on past prediction errors in order to minimize regret over time. When the semantic decoder model is uncertain, the prediction set can be intentionally large, or in some cases, may even indicate uncertainty by returning no prediction at all. This behavior prevents SemCom models from blindly producing unreliable outputs in uncertain conditions. In addition, conformal prediction provides calibrated confidence estimates that are valid regardless of the underlying model or data distribution. By avoiding unjustified certainty, it effectively addresses the issue of overconfidence.
- *Monte Carlo dropout* [12] is utilized to estimate epistemic uncertainty in neural networks, can also be applied to enhance the robustness of AI models in SemCom networks. By enabling stochastic forward passes during semantic decoding, a Monte Carlo dropout mechanism can generate multiple reconstructed data for the same received signals, forming a distribution of possible meanings. Analyzing the variation across these reconstructions allows the system to quantify its uncertainty in semantic interpretation. This not only prevents overconfident but incorrect semantic reconstructions under uncertain channel conditions but also enables the system to adjust its communication strategy dynamically based on the estimated reliability of the reconstructed meaning.

Safeguarded AI-Driven Semantic Communication Framework

In this section, we present a uniform safeguarded AI-driven SemCom framework as shown in Fig. 3. SemCom networks consist of end devices, base stations (BSs), edge servers, and cloud servers. In common SemCom systems, a semantic encoder extracts and encodes semantic information from source data based on its KB. The information is transmitted wirelessly via edge servers to the receiver. At the receiver, the semantic information are recovered by a semantic decoder. Building upon classical SemCom system architecture, our framework employs gatekeepers to monitor AI model behavior and ensure compliance with predefined safety boundaries. This theoretical foundation allows our framework to be universally deployed across any layers, tasks, and modules utilizing AI models.

Particularly, as multi-modal large language models (MLLMs) and foundation models are becoming state-of-the-art in SemCom systems, they present unique safeguarded AI challenges [13]. These large-scale models offer superior semantic understanding and cross-modal reasoning capabilities [14]. However, they simultaneously introduce significant safety concerns including complex internal representations, potential for cross-modal misalignment, and substantial computational resource consumption. Our framework can address these challenges by implementing specialized gatekeepers that monitor foundation model outputs for cross-modal semantic consistency and resource costs. Crucially, our framework focuses on controlling and validating the outputs of AI models rather than modifying the models' internal architectures or training procedures which is ideally suited for encapsulated MLLMs and foundation models.

Typically, In our proposed framework, the world model is first established as the foundation. By considering the safety-specific properties in the world model, the task-specific safety specifications are defined to evaluate the performance of AI model in SemCom networks. Gatekeepers are strategically positioned after AI models, including semantic encoders, edge servers, and semantic decoders. At each distinct deployment point, these gatekeepers serve different roles:

- After semantic encoders, gatekeepers monitor semantic representation quality and ensure KBs consistency before transmission, verifying that encoded information preserves essential semantic meaning within acceptable fidelity thresholds.
- At edge servers, gatekeepers oversee resource allocation decisions and network management operations, ensuring that semantic processing tasks are distributed safely without compromising system performance or creating bottlenecks that could lead to semantic information loss.
- Following semantic decoders, gatekeepers validate the reconstructed semantic content against original intent, checking for meaning distortion or context misinterpretation that could result in unsafe operational decisions.

Before communication begins, edge servers distribute pre-trained SemCom models with KBs to end devices. The transmitter first extracts semantic information from the source. The gatekeeper after semantic encoder evaluates the generated semantic information against established safety specifications. If they are in the safety boundaries, the gatekeeper forwards the validated output to the BS. The edge server at the BS then determines optimal network strategies (e.g., resource allocation) for transmitting the semantic information, with these decisions also subject to gatekeeper verification. Subsequently, the receiver decodes the received messages and reconstructs the original data from semantic features using its KB. The gatekeeper after semantic decoder monitors the

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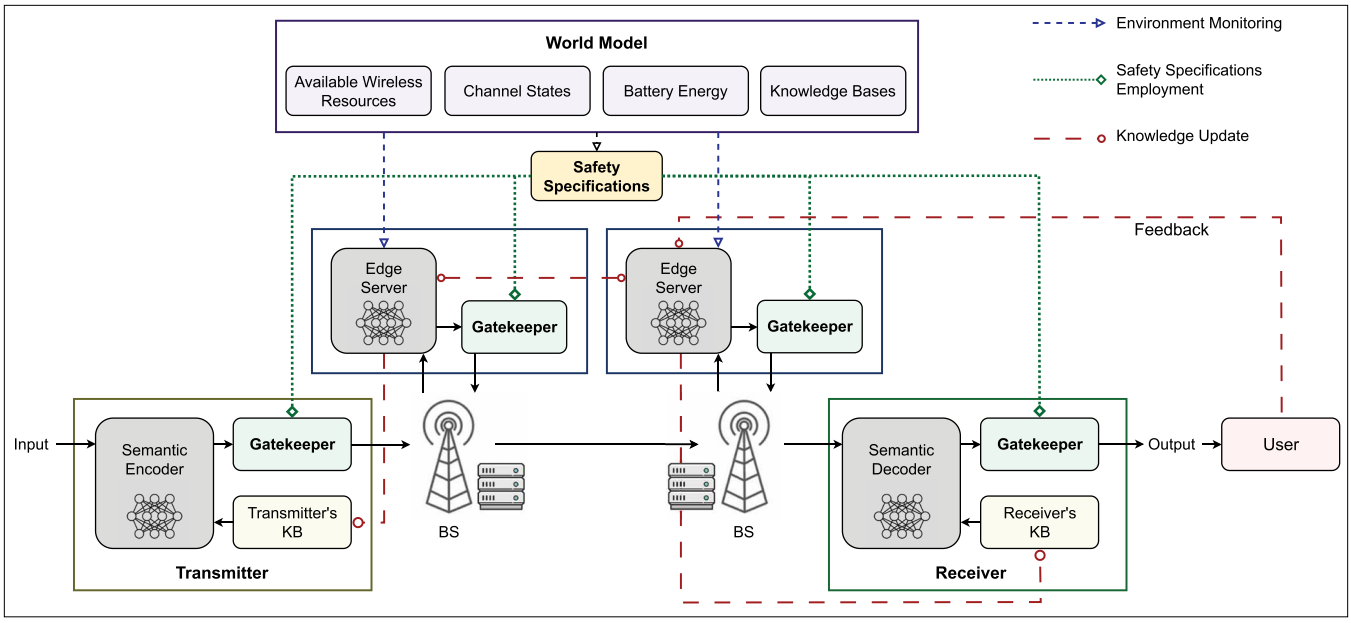

FIGURE 3. The framework of the safeguarded-AI driven SemCom framework.

reconstructed data, evaluates the entire process, and filter out unsafe output for users who will provide feedback based on gatekeepers' behavior. This feedback, along with real-time sensing data, serves as new knowledge shared among communication nodes. To maintain communication quality, knowledge updates are strategically scheduled during off-peak periods when network traffic is minimal.

Moreover, considering real-world scenarios, implementation details include:

- *Unbiased Datasets Collection:* Unbiased data collection and calibration ensures that training data for SemCom models accurately reflects real-world operational conditions. Data can be systematically gathered from controlled laboratory testbeds and open-field semantic network deployments. The controlled environment establishes baseline performance metrics by systematically varying antenna configurations, beamforming settings, and KB synchronization parameters under reproducible conditions. Conversely, real-world deployments capture the full complexity of operational scenarios, including dynamic context changes, user mobility patterns, and environmental factors.
- *Empirical Testing:* Rigorous expert testing is conducted based on comprehensive empirical understanding of both SemCom systems and practical networking environments. This testing involves multiple stages, including controlled laboratory evaluations, limited field trials, and full-scale deployments with continuous monitoring. Domain experts from SemCom systems, networking, and AI safety fields collaborate to design test scenarios that specifically target potential failure modes. The real-world evaluations incorporate adversarial testing methodologies, deliberately challenging the system with difficult edge cases to establish robust safety bounds.

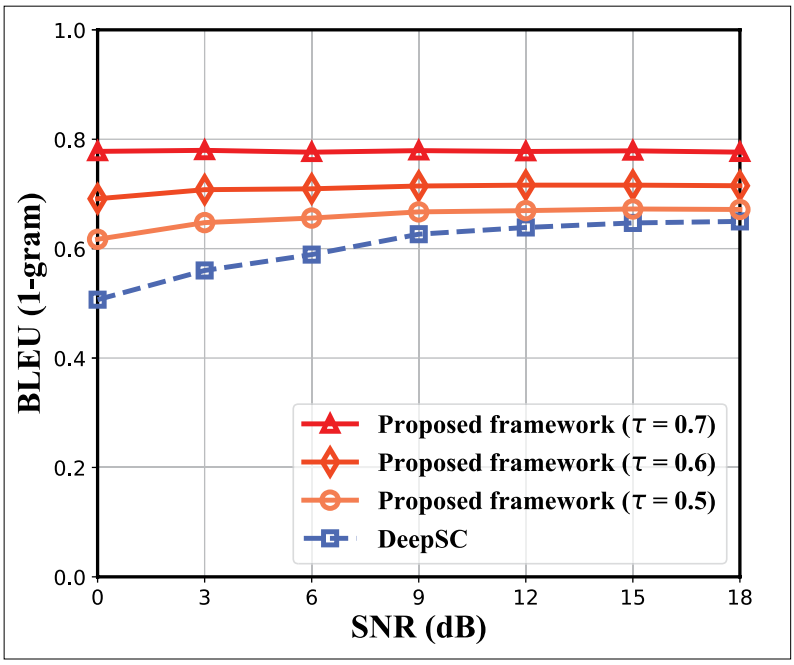

FIGURE 4. BLEU (1-gram) scores for the proposed framework with varying thresholds ($\tau = 0.5, 0.6, 0.7$) versus DeepSC.

- *Interfaces Deployment:* The mechanisms and protocols of interfaces should be designed to facilitate seamless, secure, and efficient data exchange and interaction between the safeguarded AI algorithms and the SemCom network infrastructure. These interfaces implement standardized application programming interfaces (APIs) with well-defined input/output specifications, error handling protocols, and performance monitoring capabilities. Additionally, they incorporate version compatibility layers to accommodate gradual updates of SemCom models and KBs across the network without disrupting ongoing communications.

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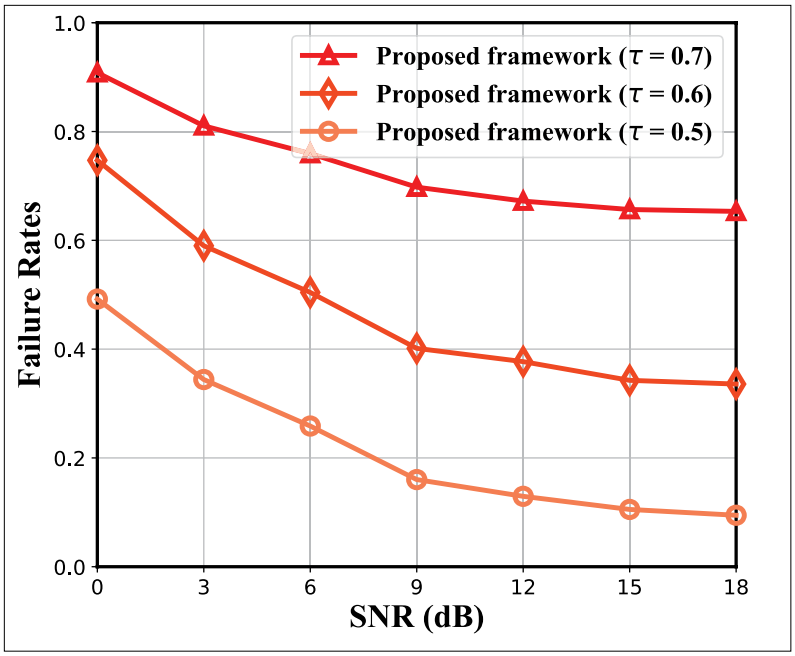

FIGURE 5. Failure rates of the proposed framework with varying thresholds ($\tau = 0.5, 0.6, 0.7$).

Case Study

In our case study, we evaluate the performance of our proposed framework through simulations by targeting DeepSC [15], a conventional SemCom AI model. We compare its performance with and without safeguarded AI implementation. To demonstrate the improvement in output's accuracy, we employ the bilingual evaluation understudy (BLEU) score, which is the same metric utilized in DeepSC. The BLEU score measures semantic similarity between transmitted and received messages by evaluating n-gram overlap and word order preservation. The gatekeeper filters out outputs with BLEU scores below predefined safety thresholds. We define threshold parameters τ at 0.5, 0.6, and 0.7 to assess diverse performances.

Fig. 4 presents the 1-gram BLEU scores of the proposed framework under different thresholds (τ = 0.5,0.6,0.7) compared to the benchmark DeepSC versus varying signal-to-noise ratio (SNR) (from 0 dB to 18 dB). Our proposed framework consistently outperforms DeepSC over the entire SNR range, especially in low-SNR conditions. As the threshold τ increases, the BLEU score of the proposed method improves, and it obtains the highest BLEU scores when τ = 0.7. Particularly, at SNR = 0 dB, the proposed framework with τ = 0.7 achieves a 54% improvement in BLEU score compared to DeepSC.

Moreover, Fig. 5 shows the failure rates of the proposed framework under different thresholds (τ = 0.5,0.6,0.7) versus varying SNR. The failure rate is calculated as the ratio of the number of filtered-out reconstructed sentences to the total number of reconstructed sentences. Our proposed framework exhibits higher failure rates at low SNRs due to poor channel conditions, particularly when τ = 0.7. As the SNR increases, the failure rate of the proposed framework gradually decreases. In contrast, as the semantic threshold increases, the failure rate rises significantly. Notably, this figure highlights a key trade-off: although the proposed framework achieves more accurate semantic reconstruction at higher thresholds, it naturally incurs a higher risk of data reconstruction failure. To balance semantic accuracy and reconstruction robustness, the gatekeeper can dynamically adjust the threshold based on channel conditions and user requirements.

Open Issues and Discussions

Although the proposed safeguarded AI-driven SemCom framework offers significant advantages, it also presents several unavoidable and complex challenges that can be addressed to fully unlock its potential.

Lack of Standardized SemCom Frameworks: For various users, SemCom lacks standardized representations of meaning, creating difficulties in defining universal safety properties. The diversity of knowledge representation approaches from neural embeddings to symbolic graphs further complicates the development of consistent verification methodologies. In these cases, safety verification should address complex properties like meaning preservation and contextual appropriateness that resist straightforward mathematical formalization. These gaps collectively hinder the development of interoperable safeguarding mechanisms that can be consistently applied across different SemCom implementations.

Resource Costs: When the gatekeepers monitor the resource costs of AI models, their own resource costs are also essential considerations for the whole SemCom network. The safeguarded AI framework introduces computational overhead that creates a recursive challenge where safety mechanisms require substantial resources to monitor resource-constrained AI models. Gatekeepers consume significant processing power for continuous behavior monitoring, real-time safety boundary evaluation, and semantic validation processes including deep contextual analysis and KB alignment and updates checks. Therefore, this framework must carefully balance the trade-off between enhanced safety assurance and computational resource consumption.

Conclusion

In this paper, we have explored enhancing AI safety for SemCom networks through safeguarded AI implementation. We begin by introducing the safeguarded AI architecture, focusing on its three core components: world model, safety specifications, and gatekeeper. Next, we have investigated detailed designs for each component to provide practical implementation guidance. Then, we have proposed our novel safeguarded AI-driven SemCom framework with a comprehensive architectural overview and implementation details. Also, we have demonstrated simulation results in a case study to evaluate the performance of our framework. Finally, we have discussed several open issues, including the lack of standardized SemCom frameworks and resource costs, which represent future research challenges in the field. We hope that this paper can pioneer the implementation of safeguarded AI in SemCom networks.

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